

Document Title

Your Name (Student ID)

September 21, 2026

1 Text Formatting

This sentence has **bold text**, *italic text*, *emphasized text*, and underlined text. SMALL CAPS and typewriter also show up occasionally.

Font sizes: scriptsize small, normal, large, Large, LARGE, huge, Huge.
Colored text in red and colored text in blue are common for highlighting contrasted statements.

2 Lists

- Primary item
- Secondary item
 - 1. Nested numbered item
 - (a) Custom-labeled sub-item
 - Dash-labeled item
 - † Symbol-labeled item (dagger)
 - ◇ Symbol-labeled item (diamond)
 - ▷ Symbol-labeled item (triangle)

(I) Roman-numbered item

(II) Another roman-numbered item

Bold-run-in label description text continues on the same line;
a manual line break is used above to keep the next line indented under it.

hello hello

hello hello

3 Math

Inline math uses $\$ \dots \$$; display math uses $\$\$ \dots \$\$, \backslash[\dots\backslash]$, or the numbered **equation** environment.

Superscripts/subscripts: x^2 , x_i , x_{i_2} (always brace nested subscripts). Greek: $\alpha, \beta, \gamma, \delta, \pi, \Pi, \sigma, \theta, \omega$. Fractions: $\frac{a}{b}$ vs. $\frac{a}{b}$ (display-style, taller); inline $\frac{a}{b}$ forces the same taller look mid-paragraph. Roots: $\sqrt{2x+1}$, $\sqrt[5]{2x+1}$. Sized brackets: $\left(\frac{1}{x^2+1}\right)$, $[\cdot]$, $\{\cdot\}$, $|\cdot|$, one-sided: $\frac{dy}{dx}\bigg|_{x=1}$. Sums/limits/integrals: $\sum_{i=1}^n x_i$, $\lim_{x \rightarrow a} f(x)$, $\int_0^\infty e^{-st} f(t) dt$. Sets/logic: $\mathbb{R}, \mathbb{Z}, \mathbb{Q}, \mathbb{N}, \forall a \in \mathbb{R}, \exists$.

$$L = I_{ij} \cdot \cos(\theta) + R^2 \tag{1}$$

$$\begin{aligned} f(x) &= x^2 + 2x + 1 \\ &= (x + 1)^2 \end{aligned} \tag{2}$$

$$x^2 + 4 = 4x \tag{3}$$

$$(x - 2)^2 = 0 \tag{4}$$

$$x = 2 \tag{5}$$

$$p(t) = \begin{cases} 0, & t < 0, \\ e^{-t}, & 0 \leq t < 3, \\ \ln(t - 2), & t \geq 3 \end{cases}$$

Equation (??) and Equation ?? are the `\eqref` and `\ref` forms respectively – `\eqref` auto-adds the parentheses.

4 Tables

Method	Accuracy	Time Complexity
Rasterization	Medium	$O(n)$
Ray Tracing	High	$O(n^2)$

Table 1: Plain bordered table

Family	Equation		Condition
	Time function	Transform	
Algebraic	1	$\frac{1}{s}$	$s > 0$
	t	$\frac{1}{s^2}$	$s > 0$

Table 2: Multirow + multicolumn table with partial cline rules

Term	Description
$f(x)$	A long description that needs to wrap onto multiple lines, hence the <code>p{5cm}</code> column type instead of <code>c</code> .

Table 3: Colored table using `xcolor[table]` (`rowcolor`/`cellcolor`)

5 Figures

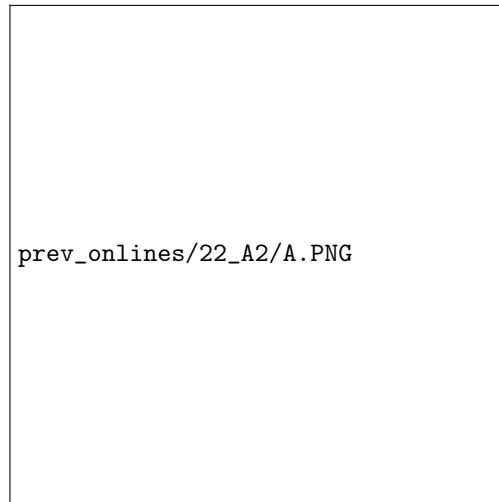
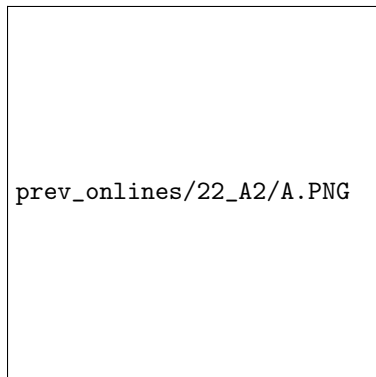
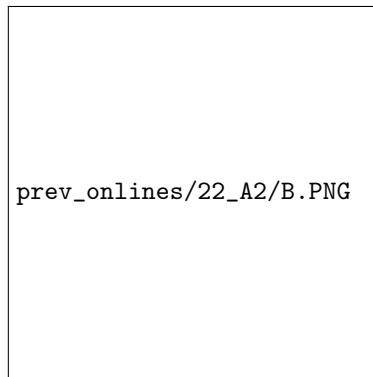


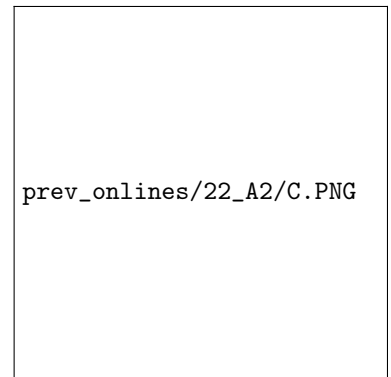
Figure 1: A single figure



(a) Output A

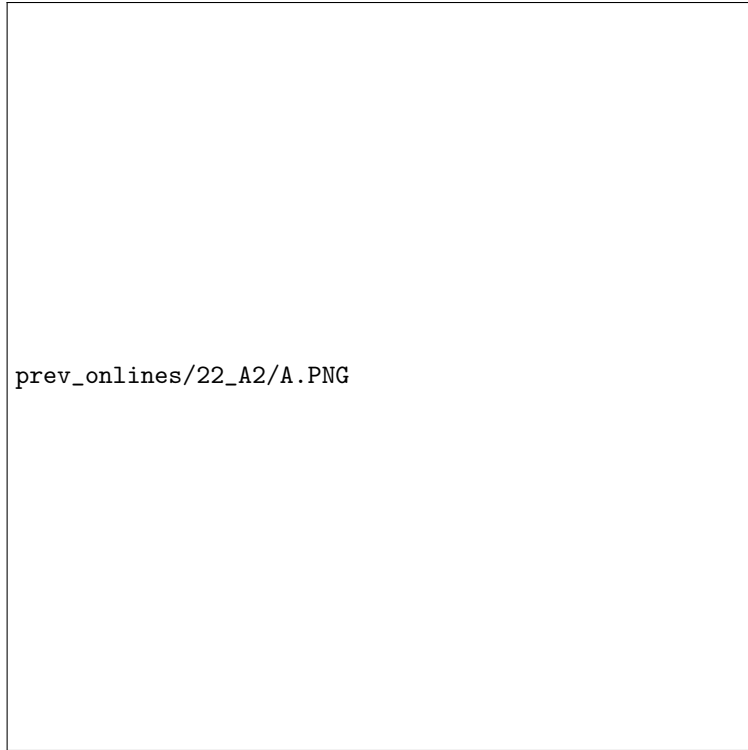


(b) Output B

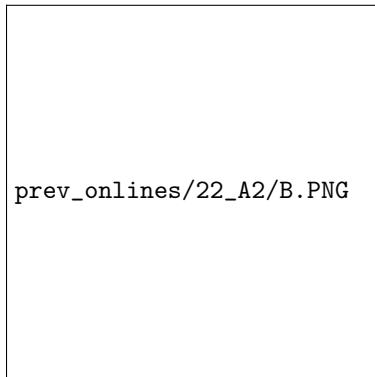


(c) Output C

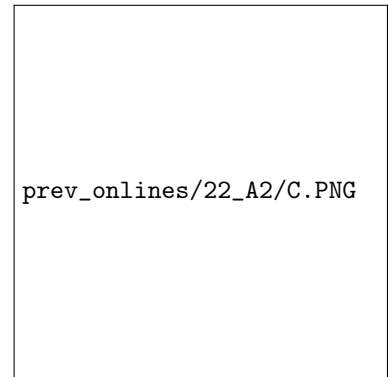
Figure 2: Three subfigures in a single row, each with its own sub-caption



(a) Main diagram



(b) Detail B



(c) Detail C

Figure 3: Asymmetric layout: one dominant figure above two smaller ones

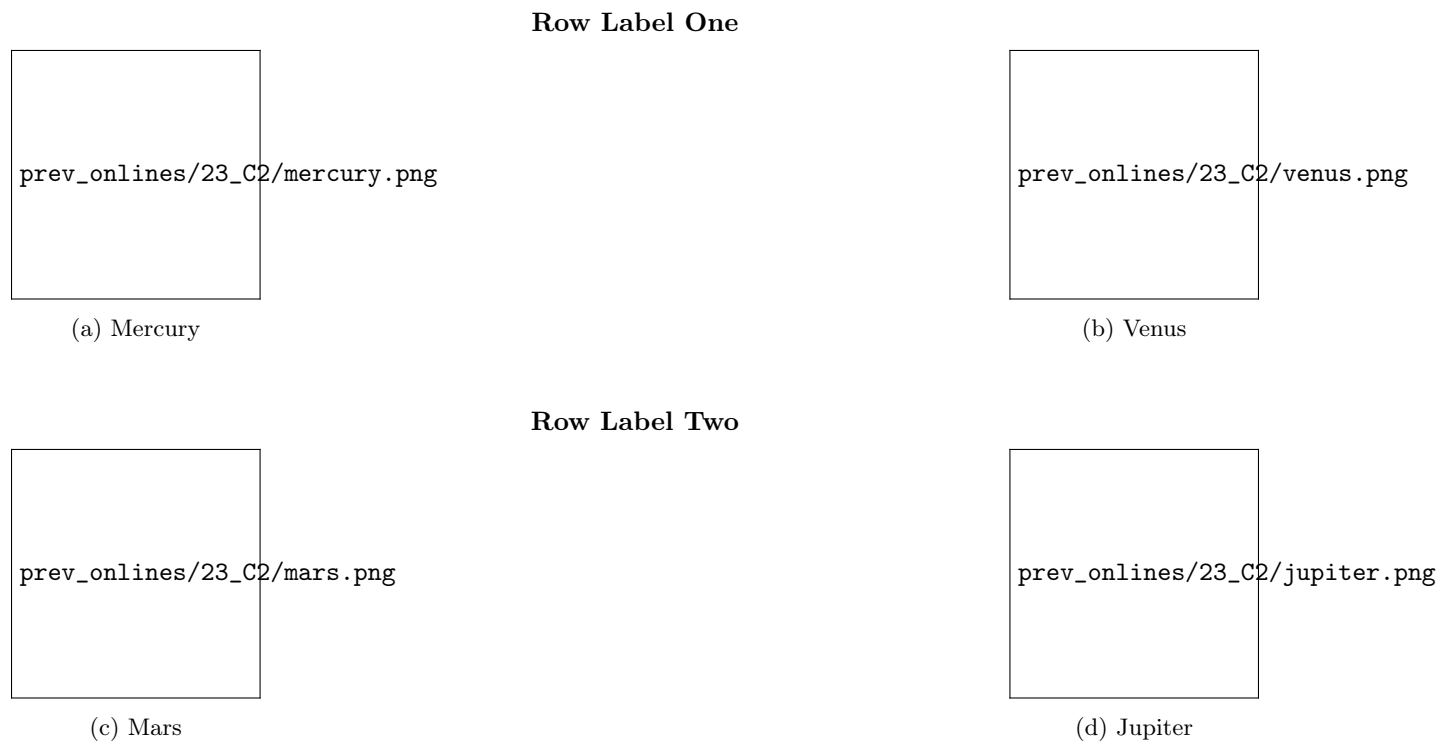


Figure 4: Grid of subfigures split into labeled row-groups

Text placed after a `wrapfigure` environment flows around the image instead of stacking above/below it, which is the key visual difference from a plain `figure`. Use `{r}` or `{l}` for the side, and a width fraction for how much horizontal space the wrap reserves. Keep enough body text below the environment to fill the wrapped height, otherwise later content can overlap the image.

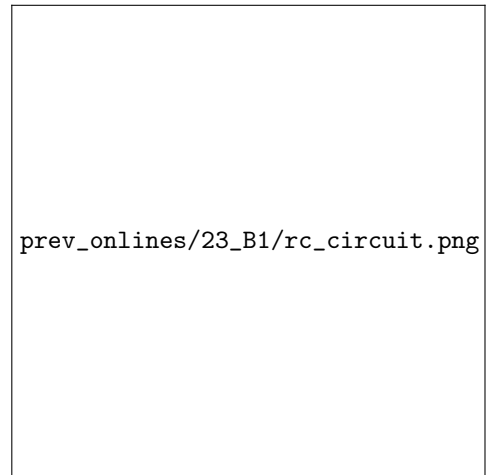


Figure 5: Wrapped figure

6 Code Listings

Listing 1: A small Python function.

```
def greet_user(name):
    # Create a greeting message using an f-string
    message = f"Hello, {name}! Welcome to Python."

    # Print the greeting message
    print(message)

    # Return the message in case it's needed elsewhere
    return message

# Example usage
user_name = "Alice"
greet_user(user_name)
```

7 Footnotes, Hyperlinks & Citations

A footnote can be attached to any word.¹ A raw link: <https://www.overleaf.com>. A labeled link: [Overleaf](#).

Citations use natbib: [?]. This resolves to a real number (not [?]) because of the `thebibliography` block near the end of this file – run `pdflatex` *twice* (or use `latexmk`) so the `.aux` data from the first pass is available to resolve it on the second.

In your actual solves, the bibliography instead comes from an external `.bib` file via `\bibliographystyle{...}` + `\bibliography{filename}` (no `.bib` extension), which needs a `bibtex`/`biber` run in addition to `pdflatex`, e.g. a file named `refs.bib` containing:

```
@book{sample2024,
  author   = {Author Name},
  title    = {Book Title},
  publisher = {Publisher},
  year     = {2024}
}
```

Common styles: `plain`, `ieeetr`, `unsrnat`, `apalike`.

8 Theorem and Proof

Proposition 1 (Sample proposition). *For a circular orbit of radius r , $\left(\frac{v}{c}\right)^2 = \frac{GM}{rc^2}$.*

Proof. The circular-orbit balance $v^2/r = GM/r^2$ gives $v^2 = GM/r$; divide by c^2 . □

See Proposition ?? for the cross-reference form of a theorem.

References

[1] Author Name. *Book Title*. Publisher, 2024.

¹This is a footnote.